COMPSCI 689
Lecture 16: Joint Probability Models

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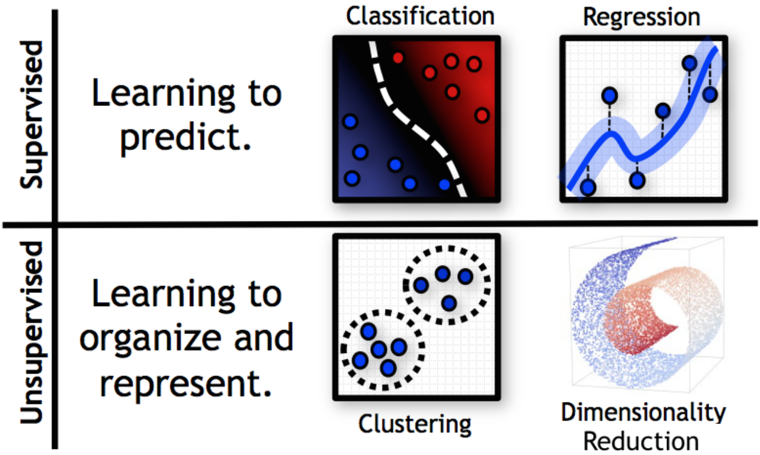
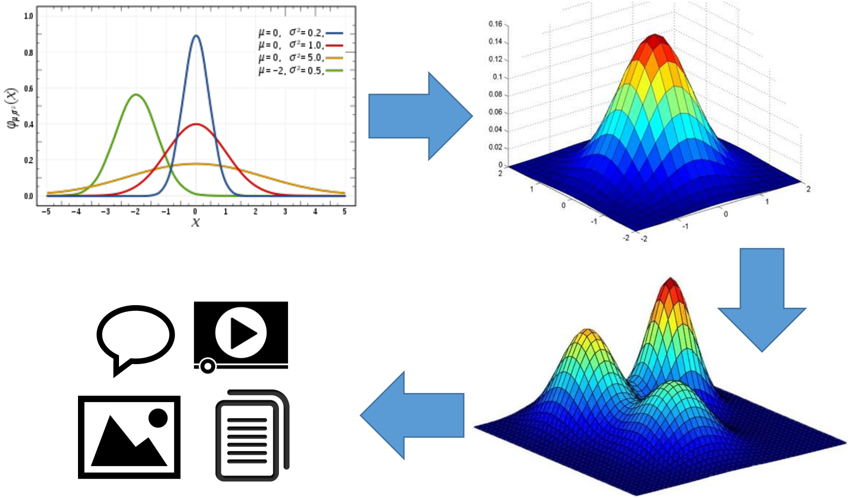
Machine Learning Tasks



Probabilistic Unsupervised Learning



Vector-Valued Random Variables

- In probabilistic unsupervised learning, our goal is to model multivariate data $\mathbf{x} = [x_1, ..., x_D]$ generated by an unknown process using a probabilistic model learned from a data set $\mathcal{D} = \{\mathbf{x}_n | 1 \leq n \leq N\}$.

- Since the data are vectors, we use vector-valued random variables to model them $\mathbf{X} = [X_1, ..., X_D]$.

- Each data dimension d takes values from a potentially different set $\mathcal{X}_d$. We have $\mathbf{x} \in \mathcal{X}$.

$$\mathcal{X} = \mathcal{X}_1 \times ... \times \mathcal{X}_D$$

Joint Distributions

- A probability distribution over a vector-valued random variable (or equivalently the joint settings of multiple random variables) is referred to as a *joint distribution*.

- When all dimensions of $\mathbf{x}$ are discrete, the joint distribution is represented by a *joint probability mass* function:

$$P(\mathbf{X} = \mathbf{x}) = P(X_1 = x_1, ..., X_D = x_d)$$

- When all dimensions of $\mathbf{x}$ are continuous, the joint distribution is represented by a *joint probability density* function:

$$p(\mathbf{X} = \mathbf{x}) = p(X_1 = x_1, ..., X_D = x_d)$$

Validity

- To be valid, a joint distribution needs to satisfy the non-negativity and normalization conditions.
- For a discrete joint distribution, we have:

$$\forall\, \mathbf{x} \in \mathcal{X}, \;\; P(\mathbf{X} = \mathbf{x}) \geq 0$$

$$\sum_{\mathbf{x} \in \mathcal{X}} P(\mathbf{X} = \mathbf{x}) = \sum_{x_1 \in \mathcal{X}_1} \cdots \sum_{x_D \in \mathcal{X}_D} P(X_1 = x_1, ..., X_D = x_D) = 1$$

Validity

- For a continuous joint distribution, we have:

$$\forall\, \mathbf{x} \in \mathcal{X}, \;\; p(\mathbf{X} = \mathbf{x}) \geq 0$$

$$\int_{\mathcal{X}} p(\mathbf{X} = \mathbf{x}) d\mathbf{x} = \int_{\mathcal{X}_1} \cdots \int_{\mathcal{X}_D} p(X_1 = x_1, ..., X_D = x_D) dx_1 \cdots dx_D = 1$$

Example: Joint Binary Distribution

x	$P(X=x \mid \theta)$
[0,0,0,0,0]	θ_0
[0,0,0,0,1]	θ_1
[0,0,0,1,0]	θ_2
[0,0,0,1,1]	θ_3
$\vdots$	
[1,1,1,1,1]	θ_{31}

Probabilistic Inference

- Joint probability distributions allow us to compute the joint probability (mass or density) of a fully specified vector of values $\mathbf{x} = [x_1, \ldots, x_D]$ of a vector valued random variable $\mathbf{X}$.
- Often, we are instead interested in computing the probability of an assignment to a subset of the dimensions of a vector-valued random variable, an operation referred to as *marginalization*.
- We can also use joint distributions to make probabilistic predictions about the distribution of one subset of dimensions given values for another subset. This operation is called *conditioning*.
- Marginalization and conditioning are the two fundamental probabilistic inference operations.

Indexing Random Variables

- Suppose we have a vector-valued random variable $\mathbf{X}$.
- Let $A \subseteq \{1, ..., D\}$ and $M = |A|$.
- We define $\mathcal{X}_A = \mathcal{X}_{A_1} \times ... \times \mathcal{X}_{A_M}$
- We define $\mathbf{X}_A = [X_{A_1}, ..., X_{A_M}]$
- We define $\mathbf{x}_A = [x_{A_1}, ..., x_{A_M}]$

Marginalization

- Suppose we have a vector-valued random variable $\mathbf{X}$. Let $A \subseteq \{1, ..., D\}$ and $B = \{1, ..., D\} \backslash A$.
- The marginal distribution of $\mathbf{X}_A$ is then given by:

$$P(\mathbf{X}_A = \mathbf{x}_A) = \sum_{\mathbf{x}_B \in \mathcal{X}_B} P(\mathbf{X}_A = \mathbf{x}_A, \mathbf{X}_B = \mathbf{x}_B)$$

$$p(\mathbf{X}_A = \mathbf{x}_A) = \int_{\mathcal{X}_B} p(\mathbf{X}_A = \mathbf{x}_A, \mathbf{X}_B = \mathbf{x}_B) d\mathbf{x}_B$$

Discrete Marginalization

$\mathbf{x}=[x_1,x_2,x_3]$	$\mathbf{P(X=x\|\theta)}$
[0,0,0]	θ_0
[0,0,1]	θ_1
[0,1,0]	θ_2
[0,1,1]	θ_3
[1,0,0]	θ_4
[1,0,1]	θ_5
[1,1,0]	θ_6
[1,1,1]	θ_7

$$
\begin{aligned}
P(X_1=1|\theta) &= P(\mathbf{X}=[1,0,0]|\theta) \\
&+ P(\mathbf{X}=[1,0,1]|\theta) \\
&+ P(\mathbf{X}=[1,1,0]|\theta) \\
&+ P(\mathbf{X}=[1,1,1]|\theta) \\
&= \theta_4 + \theta_5 + \theta_6 + \theta_7
\end{aligned}
$$

Conditioning

- Suppose we have a joint distribution on a vector-valued random variable $\mathbf{X} \in \mathbb{R}^D$. Let $A \subseteq \{1, ..., D\}$ and let $B \subseteq \{1, ..., D\} \backslash A$.

- The conditional distribution of $\mathbf{X}_A$ given $\mathbf{X}_B$ is defined as shown below:

$$P(\mathbf{X}_A = \mathbf{x}_A | \mathbf{X}_B = \mathbf{x}_B) = \frac{P(\mathbf{X}_A = \mathbf{x}_A, \mathbf{X}_B = \mathbf{x}_B)}{P(\mathbf{X}_B = \mathbf{x}_B)}$$

$$p(\mathbf{X}_A = \mathbf{x}_A | \mathbf{X}_B = \mathbf{x}_B) = \frac{p(\mathbf{X}_A = \mathbf{x}_A, \mathbf{X}_B = \mathbf{x}_B)}{p(\mathbf{X}_B = \mathbf{x}_B)}$$

Discrete Conditioning

$\mathbf{x}=[x_1,x_2,x_3]$	$P(X=x\|\theta)$
[0,0,0]	θ_0
[0,0,1]	θ_1
[0,1,0]	θ_2
[0,1,1]	θ_3
[1,0,0]	θ_4
[1,0,1]	θ_5
[1,1,0]	θ_6
[1,1,1]	θ_7

$$P(X_2=0, X_3=0 \mid X_1=1, \theta)$$
$$= P(\mathbf{X}=[1,0,0] \mid \theta) \,/\, P(X_1=1 \mid \theta)$$
$$= \theta_4 \,/\, (\theta_4 + \theta_5 + \theta_6 + \theta_7)$$

Multivariate Normal

- The multivariate normal (or Gaussian) distribution is a fundamental building block for unsupervised learning with vector-valued random variables $\mathbf{X} \in \mathbb{R}^D$.

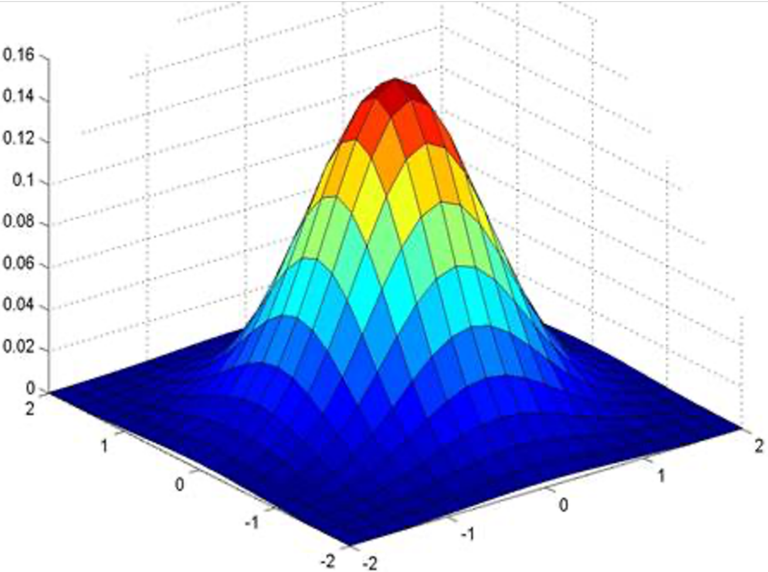
- The distribution has two parameters $\theta = [\mu, \Sigma]$. μ is the mean vector and Σ is the covariance matrix.

- The probability density is given below (assuming $\mathbf{x}$ and μ are column vectors):

$$\mathcal{N}(\mathbf{x}; \mu, \Sigma) = \frac{1}{|2\pi\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\right)$$

- We have $\mu \in \mathbb{R}^D$ and $\Sigma \in \mathbb{S}_+^D$, the space of symmetric, positive definite $D \times D$ matrices.

Multivariate Normal



Marginalization for MVNs

- The multivariate normal distribution has the remarkable (and convenient) property of being closed under marginalization.

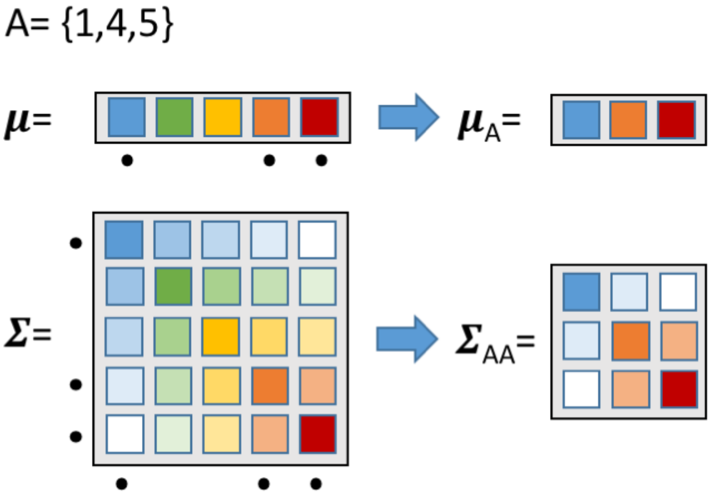
- Suppose we have an MVN $p(\mathbf{X}|\theta) = \mathcal{N}(\mathbf{X}; \mu, \Sigma)$ for $\mathbf{X} \in \mathbb{R}^D$. Let $A \subseteq \{1, ..., D\}$, $B = \{1, ..., D\}/A$, and $M = |A|$. We have:

$$p(\mathbf{X}_A = \mathbf{x}_A) = \mathcal{N}(\mu_A, \Sigma_{AA})$$

  where $\mu_A = [\mu_{A_1}, ..., \mu_{A_M}]$ and $(\Sigma_{AA})_{ij} = \Sigma_{A_i, A_j}$.

- In other words, we get the marginal distribution on a subset of $\mathbf{X}$ just by discarding the elements of μ that correspond to B, and the rows and columns of Σ that correspond to B.

Marginalization for MVNs: Example



Conditioning for MVNs

- The multivariate normal distribution has the remarkable (and convenient) property of also being closed under conditioning.
- Suppose we have an MVN $p(\mathbf{X}|\theta) = \mathcal{N}(\mathbf{X}; \mu, \Sigma)$ for $\mathbf{X} \in \mathbb{R}^D$. Let $A \subseteq \{1, ..., D\}$, $B = \{1, ..., D\} \backslash A$. We have:

$$p(\mathbf{X}_A = \mathbf{x}_A | \mathbf{X}_B = \mathbf{x}_B) = \mathcal{N}(\mathbf{x}_A; \mu_{A|B}, \Sigma_{AA|B})$$

$$\mu_{A|B} = \mu_A + \Sigma_{AB}(\Sigma_{BB})^{-1}(\mathbf{x}_B - \mu_B)$$

$$\Sigma_{AA|B} = \Sigma_{AA} - \Sigma_{AB}(\Sigma_{BB})^{-1}\Sigma_{BA}$$

Conditioning for MVNs: Example